

Toward Reproducible Desk-Based Vision Experiments: A Multi-Camera Study of Papers, Pens, and Other Obstacles

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Abstract—Reproducing a video experiment requires more than sharing a model checkpoint. We define a compact scene manifest that records cameras, object placement, synchronization, and expected occlusions. Three synthetic camera views reveal how a misplaced pen changes object visibility and event timing. A small ablation shows that explicit scene metadata makes discrepancies easier to diagnose.

Keywords—video analysis; temporal reasoning; reproducibility; synthetic experiments.

1 Introduction

A desk can seem like a controlled environment until a pen covers the object of interest. Small layout changes alter visibility, and unsynchronized cameras can disagree about event order. We propose a minimal record of these conditions to support repeatable experiments.

2 Related Work

Fictional reproducibility checklists recommend saving capture settings [1]. Multiview studies document the cost of clock mismatch [2]; desk-layout studies document occlusion sensitivity [3]. We combine these concerns in a single inspectable manifest.

3 Methodology

Each recording has a YAML manifest containing camera identifiers, nominal frame rates, timestamp origin, object coordinates, and planned occluders. Frames retain native timestamps; alignment uses nearest timestamp with a 20 ms tolerance. We report unmatched frames instead of silently duplicating neighbors.

4 Experimental Setup

We rendered 45 trials with three cameras at 25 frames/s. Trials differ only in pen placement, paper rotation, and a simulated 0–40 ms camera offset. The task is to identify when a colored card is placed on the desk. One camera has a deliberate blind region.

5 Manifest Schema

The manifest separates recorded facts from intended settings. A camera’s nominal frame rate

is a setting; every frame’s timestamp is an observation. Object coordinates identify the layout at capture start, and a change log records deliberate movement thereafter. Each clip also names its calibration file and the renderer seed. This separation prevents an intended 25 frames/s stream from being mistaken for proof that all frames were delivered.

For synchronization we convert device timestamps to a common origin, then match each frame to the nearest frame in the other view within tolerance. Unmatched frames remain unmatched. Repeating the previous image would create a visually smooth sequence while changing the apparent duration of an action. We retain both raw and aligned indices so that an analyst can trace an event back to the acquired frame.

6 Audit Procedure

A lightweight validator checks unique camera IDs, monotonic timestamps, declared image dimensions, and the existence of referenced calibration files. It also computes the achieved frame interval distribution. Scene coordinates are checked against image bounds where applicable. The validator emits a short report before any model evaluation, which makes capture problems visible while they can still be corrected. The benchmark includes one deliberately duplicated timestamp to show that this check produces an explicit error rather than an arbitrary alignment.

7 Results

Without timestamp checks, the largest apparent cross-view onset discrepancy was 80 ms. Applying the manifest and rejecting unmatched frames reduced the maximum among retained pairs to 20 ms. Moving one pen by 4 cm raised occlusion in camera B from 3% to 18%, explaining most missed card onsets in that view.

Configuration	Offset	Occlusion B
Original placement	0–40 ms	3%
Shifted pen	0–40 ms	18%

Controlled scene perturbation.

8 Discussion

A manifest does not eliminate variation; it identifies which variation was intentional. The experiment uses synthetic cameras and perfect object coordinates, so the diagnostic benefit will likely be smaller with uncertain real-world calibration. Publishing the renderer seed is also necessary for exact regeneration.

9 Ablation and Diagnosis

We repeated alignment without the offset field, without the pen-placement log, and with both omitted. Without the offset, event onset could appear one or two frames apart even when the same renderer event drove all cameras. Without the placement log, the rise in camera-B misses appeared to be a model regression. Restoring both fields allowed the discrepancy to be attributed to timing and visibility before any detector threshold was changed.

The 20 ms tolerance does not guarantee 20 ms event accuracy. It only bounds the timestamp difference of retained frame pairs. Exposure timing, rolling shutter, and event annotation can add independent uncertainty. A real experiment should record the clock synchronization method and, where possible, image a shared temporal cue. We also recommend listing excluded frames and the rule that excluded them; a cleaned sequence by itself cannot reveal the cost of cleaning.

10 Reuse Scenario

A second researcher can reproduce the desk layout using the manifest, run the validator, and compare the resulting visibility summary with the original. Exact object placement will still differ, so the goal is a documented comparison rather than a claim of identical video. A short photograph of the setup and a versioned calibration target would strengthen this record. The same approach extends to other tabletop tasks when the object list and event definitions are changed.

11 Dataset Release Checklist

A reusable package should contain the original frames, timestamp tables, calibration inputs, manifest, validation report, and scripts that generated the aligned view. Checksums can establish which files were used for a published result. A small example clip should accompany the schema so that another group can verify its parser before processing the complete dataset. Any manual correction to an event label belongs in a versioned annotation log with the reason and reviewer recorded.

12 Conclusions and Future Work

Scene metadata helps explain timing and visibility differences across cameras. Future work will validate the manifest on real recordings and add machine-checkable provenance for annotations and dropped frames.

References

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